VHDL NOT gate Simulation Report

Course: 디지털논리회로설계
Assignment: VHDL NOT Gate Simulation
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1. Objective

The purpose of this assignment is to design a NOT gate using VHDL and simulate it in Logisim-evolution using the Questa simulator.

The NOT gate produces the complement of the input signal.

$$y = \overline{a}$$

The expected operation is

Input a	Output y
0	1
1	0

2. VHDL Code and Circuit

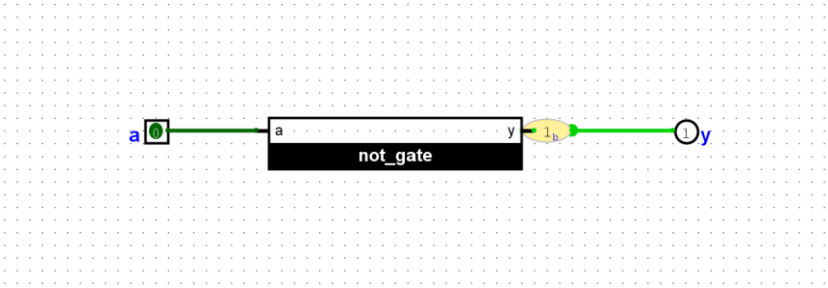
The following VHDL code defines the operation of the NOT gate.

```vhdl
library ieee;
use ieee.std_logic_1164.all;

entity not_gate is
    port (
        a : in std_logic;
        y : out std_logic
    );
end not_gate;

architecture behavior of not_gate is
begin
    y <= not a;
end behavior;
```

The NOT gate VHDL code was imported into Logisim-evolution and connected between the input a and the output y.

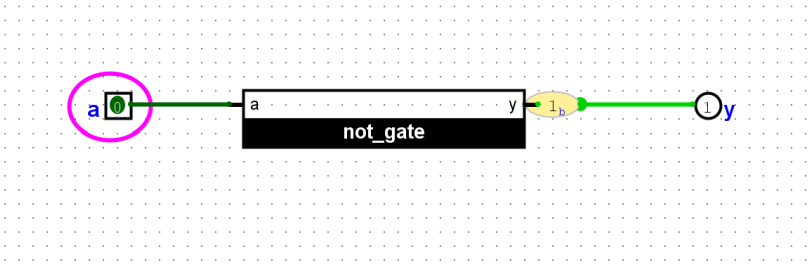


3. Simulation Results

Case 1: $a = 0$

When the input a was set to 0, the output y became 1.

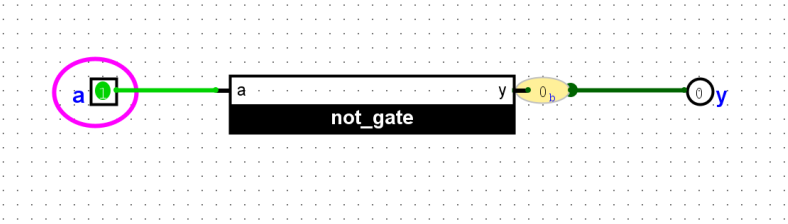
$$a = 0 \rightarrow y = 1$$



Case 2: $a = 1$

When the input a was set to 1, the output y became 0.

$$a = 1 \rightarrow y = 0$$



4. Conclusion

The Not gate was successfully implemented using VHDL and simulated in Logisim-evolution with the Questa simulator.

The simulation results were:

$$a = 0 \rightarrow y = 1$$

$$a = 1 \rightarrow y = 0$$

Therefore, the VHDL circuit correctly performed the logical NOT operation.